

REDDING, CA, USA

WMO: 725920

Lat: 40.518N

Lon: 122.299W

Elev: 152

StdP: 99.52

Time Zone: -8.00 (NAP)

Period: 94-19

WBAN: 24257

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	-1.5	-0.1	-11.6	1.4	12.9	-9.2	1.8	13.2	15.3	11.0	12.8	11.5	0.5	30	0.459

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	17.2	40.8	20.0	39.1	19.5	37.4	18.9	21.9	35.8	21.1	34.7	20.3	33.7	3.5	160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
17.5	12.8	26.3	16.3	11.8	25.7	15.3	11.0	25.0	64.4	36.3	61.4	34.5	58.8	33.9	24.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 11.4	8.9	7.5	DB	-4.2	44.0	1.8	1.5	-5.5	45.0	-6.5	45.9	-7.5	46.7	-8.8	47.8	(4)
(5)			WB	-5.0	23.4	1.7	0.9	-6.3	24.0	-7.3	24.5	-8.3	25.1	-9.6	25.7	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	17.3	8.6	10.0	12.5	15.1	19.8	24.9	28.3	27.0	23.8	17.9	11.5	8.1	(6)	
	DBStd	7.92	3.10	3.36	3.64	4.11	4.33	4.35	2.96	2.73	3.54	3.70	3.57	3.09	(7)	
	HDD10.0	222	64	37	16	5	0	0	0	0	0	0	24	75	(8)	
	HDD18.3	1455	303	233	186	115	35	4	0	0	3	53	206	317	(9)	
	CDD10.0	2899	21	38	92	157	306	447	568	527	415	244	70	16	(10)	
	CDD18.3	1089	0	1	3	17	82	201	310	268	168	39	1	0	(11)	
	CDH23.3	15368	3	9	69	281	1100	2759	4491	3754	2266	597	38	1	(12)	
	CDH26.7	8855	0	1	11	89	502	1584	2846	2312	1277	227	6	0	(13)	
(14) Wind	WSAvg	2.6	2.3	2.9	3.0	2.9	2.9	3.0	2.5	2.3	2.3	2.5	2.1	2.4	(14)	
(15) Precipitation	PrecAvg	963	178	146	132	64	41	18	3	9	23	50	132	167	(15)	
	PrecMax	1857	686	401	416	233	195	65	84	139	189	223	412	449	(16)	
	PrecMin	510	8	4	13	0	0	0	0	0	0	0	7	0	(17)	
	PrecStd	281	134	103	94	52	48	18	13	22	37	48	99	117	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	23.4	24.9	28.2	32.1	37.1	42.3	43.5	42.4	40.2	35.1	27.3	20.7	(19)	
		MCWB	11.2	13.3	14.5	16.1	18.4	20.3	20.9	20.6	18.6	16.9	14.3	10.2	(20)	
	2%	DB	19.9	21.7	25.3	29.3	34.3	39.2	41.1	40.0	37.7	31.6	23.9	17.7	(21)	
		MCWB	10.1	11.6	13.9	15.4	18.0	19.6	20.4	19.5	17.8	16.0	13.4	9.4	(22)	
	5%	DB	16.8	19.1	22.8	26.8	32.0	36.9	39.4	38.0	35.8	29.2	21.2	15.4	(23)	
		MCWB	9.4	10.6	12.8	14.8	17.0	18.7	19.9	19.0	17.6	15.0	12.4	9.3	(24)	
	10%	DB	14.3	16.4	20.2	23.9	29.5	34.6	37.6	36.4	33.9	26.7	18.4	13.6	(25)	
		MCWB	8.9	9.8	11.7	13.7	16.1	18.0	19.4	18.5	17.2	14.2	11.1	9.3	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	13.5	14.6	16.1	18.3	20.7	22.6	23.1	22.6	21.7	18.3	16.2	13.9	(27)	
		MCDB	16.8	20.0	24.1	28.0	31.7	35.5	37.2	38.1	32.8	30.5	23.7	15.2	(28)	
	2%	WB	12.1	13.5	14.9	16.7	19.1	21.1	22.1	21.3	19.9	17.2	14.8	12.5	(29)	
		MCDB	15.1	18.3	22.5	26.3	30.9	34.2	36.7	35.2	32.9	28.6	20.8	14.0	(30)	
	5%	WB	11.1	12.2	13.8	15.5	18.1	20.1	21.3	20.3	18.8	16.2	13.6	11.5	(31)	
		MCDB	14.1	16.7	20.4	24.5	29.5	33.4	35.6	34.0	32.0	26.3	18.5	13.1	(32)	
	10%	WB	10.3	11.0	12.7	14.4	17.1	19.1	20.6	19.6	17.9	15.2	12.6	10.3	(33)	
		MCDB	13.5	14.9	18.5	22.3	27.5	32.0	34.4	33.0	31.3	24.1	16.7	12.6	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	9.9	10.3	11.5	12.7	14.0	15.3	17.2	17.9	17.3	14.8	11.5	9.3	(35)	
		MCDBR	16.5	15.8	16.4	17.5	18.1	18.1	19.3	19.9	20.6	19.2	17.0	14.0	(36)	
	5% WB	MCWBR	8.7	8.1	7.7	7.2	6.0	5.3	5.3	5.7	6.6	7.5	8.3	7.9	(37)	
		MCWBR	11.0	12.1	14.1	15.9	16.6	17.1	17.7	18.1	18.0	16.7	13.5	7.7	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.313	0.319	0.329	0.347	0.351	0.350	0.354	0.341	0.339	0.329	0.309	0.314	(40)	
		taud	2.535	2.524	2.508	2.443	2.471	2.488	2.473	2.500	2.494	2.533	2.566	2.528	(41)	
	Ebn at Noon		854	901	927	926	926	922	916	921	903	876	853	820	(42)	
		Edh at Noon	73	85	96	109	109	107	108	102	96	83	70	69	(43)	
(44) All-Sky Solar Radiation	RadAvg		2.02	2.92	4.14	5.68	7.04	8.05	8.19	7.28	5.80	3.92	2.37	1.70	(44)	
	RadStd		0.39	0.42	0.55	0.59	0.52	0.46	0.26	0.26	0.24	0.35	0.27	0.36	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	+0.51	N/A	-1.48	N/A	N/A	N/A	-37	-93	+166	+102
(47)	Regional (16 neighbors)	N/A	N/A	N/A	+0.42	N/A	N/A	N/A	N/A	+102	+26

Nomenclature: See separate page